

ALBERT HALL AGCM, AB, Canada

WMO: 710050

Lat: 53.1134N

Lon: 111.1812W

Elev: 2159

StdP: 13.59

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.0	-21.6	-33.4	0.9	-27.0	-27.4	1.3	-21.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	83.6	63.7	80.3	62.2	77.3	61.2	67.5	78.5	65.3	76.2	63.3	73.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	94.9	73.3	61.2	87.5	70.6	59.2	81.4	68.4	33.2	78.7	31.4	76.0	29.8	73.1	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-35.4	89.5	4.5	3.7	-38.6	92.2	-41.2	94.3	-43.7	96.4	-46.9	99.1		
(5)			-36.0	72.0	4.5	1.9	-39.2	73.4	-41.9	74.5	-44.4	75.5	-47.6	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	10.4	8.6	21.8	37.0	50.9	58.8	62.9	60.2	51.5	38.5	22.4	9.2		
	DBStd	22.59	15.06	14.49	13.64	9.18	7.74	5.21	4.63	5.36	7.87	8.55	12.16	14.26		
	HDD50	6281	1227	1159	874	396	84	5	0	2	74	367	828	1265		
	HDD65	10577	1692	1579	1339	840	438	194	96	162	407	822	1278	1730		
	CDD50	1239	0	0	0	7	113	270	401	319	119	10	0	0		
	CDD65	60	0	0	0	0	1	9	33	15	2	0	0	0		
	CDH74	1479	0	0	0	4	135	188	545	401	202	4	0	0		
	CDH80	324	0	0	0	0	18	27	139	81	59	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.3	0.7	0.5	0.6	1.2	1.7	2.8	3.0	2.3	1.3	0.7	0.7	0.7		
(15) Precipitation	PrecMax	22.1	2.4	0.9	1.5	4.1	3.7	5.5	6.5	4.6	3.1	1.8	3.0	1.3		
	PrecMin	8.2	0.2	0.1	0.2	0.1	0.2	0.6	1.0	0.5	0.0	0.1	0.1	0.2		
	PrecStd	3.1	0.5	0.2	0.3	0.8	0.9	1.1	1.2	0.9	0.8	0.4	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	41.0	54.2	71.5	82.6	82.8	88.8	86.4	86.7	71.9	55.0	41.3		
		MCWB	36.6	36.7	43.1	50.0	56.6	62.8	67.7	63.8	60.7	54.4	44.2	35.5		
	2%	DB	39.3	37.0	47.5	64.9	78.1	79.0	83.7	81.8	79.9	62.7	45.8	37.0		
		MCWB	34.9	33.3	39.6	47.2	54.8	60.3	67.0	63.7	58.4	49.4	38.8	32.6		
	5%	DB	36.2	34.2	42.3	58.6	73.0	75.6	79.8	78.6	74.3	57.5	41.2	33.3		
		MCWB	32.7	31.6	36.9	44.3	52.9	59.4	65.2	63.4	56.6	46.1	36.2	29.9		
	10%	DB	32.1	31.0	38.5	53.5	68.7	72.1	76.2	74.9	68.4	52.8	37.6	29.2		
		MCWB	29.5	28.7	34.3	41.7	50.9	57.5	63.5	61.5	54.3	43.9	34.0	26.9		
	0.4%	WB	37.3	36.8	44.9	51.3	58.5	66.1	71.6	69.5	62.8	55.0	44.9	36.0		
		MCDB	41.3	40.5	53.2	71.0	77.3	77.0	83.3	79.4	78.6	69.4	52.8	40.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	33.9	40.1	47.7	56.1	63.2	68.6	66.6	59.9	51.0	39.8	33.1		
		MCDB	39.1	36.1	46.7	62.7	72.7	74.3	79.9	77.3	76.9	59.4	44.5	36.7		
	5%	WB	32.8	31.8	37.1	44.9	54.2	61.2	66.5	64.3	57.4	47.8	36.8	30.1		
		MCDB	36.0	34.3	42.0	56.9	70.2	71.5	77.8	75.5	71.7	56.3	40.6	32.9		
	10%	WB	29.5	28.6	34.6	42.4	52.4	59.2	64.5	62.3	55.2	44.8	34.1	26.9		
		MCDB	32.2	31.0	38.2	52.3	66.8	69.0	74.2	72.3	67.4	52.6	37.3	29.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	20.5	19.2	21.2	25.5	21.6	22.5	24.5	26.2	21.4	16.8	18.0		
		MCDBR	20.8	20.9	22.7	32.4	33.7	28.3	30.0	32.8	38.5	30.8	21.5	21.6		
		MCWBR	17.2	17.8	16.2	17.0	14.4	12.5	15.6	16.5	18.2	17.8	15.0	18.2		
		MCDBR	20.3	20.4	22.1	30.0	29.9	24.1	27.5	28.4	34.3	29.1	20.5	21.4		
	5% WB	MCWBR	17.1	17.6	16.0	16.3	13.5	12.1	15.0	15.5	17.1	17.4	15.0	18.4		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.292	0.294	0.333	0.353	0.372	0.365	0.379	0.315	0.279	0.252	0.226		
	taud		2.317	2.224	2.371	2.355	2.348	2.328	2.371	2.312	2.464	2.541	2.448	2.362		
	Ebn at Noon		239	257	286	285	284	279	278	267	275	267	241	232		
	Edh at Noon		18	28	30	35	37	38	36	36	27	20	16	14		
(44) All-Sky Solar Radiation	RadAvg		325	660	1133	1495	1844	1889	1970	1605	1131	662	325	242		
	RadStd		32	49	88	135	127	135	103	123	121	91	34	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page